

Building a Library Management System using GitHub Copilot

We'll build a simple **web-based Library Management System** using:

- HTML
 - CSS
 - JavaScript
 - GitHub Copilot inside VS Code
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Guided Practice

Building a Library Management System using GitHub Copilot

Objective

Create a small system that can:

- Add books
- Display books
- Delete books

And observe how GitHub Copilot assists with code generation.

Project Structure

Create a folder:

```
library-system/  
├── index.html  
├── style.css  
└── script.js
```

Open this folder in **VS Code** with Copilot enabled.

Step 1: Generate HTML Layout with Copilot

Open index.html and type:

```
<!-- Create a simple Library Management System UI with form and table -->
```

Press:

Ctrl + Enter

Copilot Suggests (Accept with TAB)

```
<!DOCTYPE html>
<html>
<head>
  <title>Library Management System</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<h1>Library Management System</h1>

<form id="bookForm">
  <input type="text" id="title" placeholder="Book Title" required>
  <input type="text" id="author" placeholder="Author" required>
  <input type="number" id="year" placeholder="Year" required>
  <button type="submit">Add Book</button>
</form>

<table>
  <thead>
    <tr>
      <th>Title</th>
      <th>Author</th>
      <th>Year</th>
      <th>Action</th>
    </tr>
  </thead>
  <tbody id="bookList"></tbody>
</table>

<script src="script.js"></script>
</body>
</html>
```

Step 2: Generate CSS Styling

Open `style.css` and type:

```
/* Style Library Management System */
```

Copilot suggests:

```
body {  
  font-family: Arial;  
  text-align: center;  
}  
  
form {  
  margin-bottom: 20px;  
}  
  
input {  
  padding: 8px;  
  margin: 5px;  
}  
  
button {  
  padding: 8px 15px;  
  cursor: pointer;  
}  
  
table {  
  margin: auto;  
  border-collapse: collapse;  
  width: 60%;  
}  
  
th, td {  
  border: 1px solid #333;  
  padding: 8px;  
}
```

Step 3: Generate JavaScript Logic

Open `script.js` and type:

```
// Add book management functionality
```

Copilot suggests:

```
const form = document.getElementById("bookForm");
const bookList = document.getElementById("bookList");

form.addEventListener("submit", function(e) {
  e.preventDefault();

  const title = document.getElementById("title").value;
  const author = document.getElementById("author").value;
  const year = document.getElementById("year").value;

  const row = document.createElement("tr");

  row.innerHTML = `
    <td>${title}</td>
    <td>${author}</td>
    <td>${year}</td>
    <td><button class="delete">Delete</button></td>
  `;

  bookList.appendChild(row);
  form.reset();
});

bookList.addEventListener("click", function(e) {
  if (e.target.classList.contains("delete")) {
    e.target.parentElement.parentElement.remove();
  }
});
```

Step 4: Run the Application

Open `index.html` in browser.

You can now:

- ✓ Add books
- ✓ See them in table
- ✓ Delete books

Copilot in Action (What You Observed)

- Generated UI layout
- Suggested event handling
- Created table rows
- Handled delete logic

All with minimal typing.

Enhancements (Ask Copilot)

Try typing comments:

```
// Save books in local storage
// Add search functionality
// Add edit book option
```

Copilot will generate enhancements.

Quick Check

Q1. What role does GitHub Copilot play?

✓ AI pair programmer

Q2. Which languages used?

✓ HTML, CSS, JavaScript

Q3. True/False

Copilot replaces developers

☐ False

Q4. Name one feature you built

✓ Add/Delete books

Summary

This guided practice demonstrates how GitHub Copilot:

- Speeds up coding
 - Reduces boilerplate
 - Helps beginners and professionals
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